C1 Atomic Structure and the Periodic Table – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Atoms, Elements and Mixtures **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Atom				Smallest part of an element that can exist
2	Element				Substance made of only one type of atom
3	Compound				Two or more elements chemically combined in fixed proportions
4	Mixture				Two or more substances not chemically combined
5	Methods to separate mixtures				Filtration, crystallisation, distillation, chromatography
6	Radius of an atom				About 0.1 nm (1×10^{-10} m)
7	Size of the nucleus compared with the atom				About 1/10 000 of the radius
8	Relative masses of proton, neutron, electron				1, 1, very small
9	Relative charges of proton, neutron, electron				+1, 0, −1
10	Why atoms have no overall charge				Equal numbers of protons and electrons

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 2: Atomic Number, Isotopes and the Atomic Model **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Atomic number				Number of protons
2	Mass number				Number of protons + neutrons
3	Isotopes				Same number of protons, different number of neutrons
4	Relative atomic mass				Average mass of an element's atoms, taking isotope abundance into account
5	Plum pudding model				Ball of positive charge with electrons embedded in it
6	Experiment that replaced the plum pudding model				Alpha particle scattering
7	Conclusion of alpha scattering				Mass is concentrated in a tiny positive nucleus
8	Scientist who put electrons in shells				Niels Bohr
9	Scientist who showed neutrons exist				James Chadwick
10	Maximum electrons in the first three shells				2, 8, 8

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 3: The Periodic Table **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Modern periodic table is ordered by				Atomic number
2	Group number tells you				Number of outer electrons
3	Period number tells you				Number of electron shells
4	Ions formed by metals				Positive ions
5	Ions formed by non-metals				Negative ions
6	Position of metals in the periodic table				Left and centre
7	Mendeleev ordered elements by				Atomic weight and properties
8	Why Mendeleev left gaps				For undiscovered elements
9	Why Mendeleev swapped some pairs				To match their properties
10	What explained the swapped pairs				Isotopes

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 4: Groups 0, 1 and 7 **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Name of Group 0				Noble gases
2	Why noble gases are unreactive				Full outer shell (stable)
3	Trend in boiling point down Group 0				Increases
4	Name of Group 1				Alkali metals
5	Trend in reactivity down Group 1				Increases
6	Products of a Group 1 metal with water				Metal hydroxide + hydrogen
7	Equation for sodium with water				$2Na + 2H_2O \rightarrow 2NaOH + H_2$
8	Name of Group 7				Halogens
9	Trend in reactivity down Group 7				Decreases
10	Trend in melting and boiling point down Group 7				Increases

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA GCSE Chemistry • Set 5: Group 7 and Reactivity Paper 1

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Form halogens exist in				Diatomic molecules
2	Appearance of chlorine				Green gas
3	Appearance of bromine				Red-brown liquid
4	Appearance of iodine				Grey-black solid
5	Halogen displacement				More reactive halogen displaces a less reactive one from its salt
6	Products of chlorine + potassium bromide				Potassium chloride + bromine
7	Charge on Group 1 ions				1+
8	Charge on Group 7 ions				1−
9	Why Group 1 gets more reactive down the group				Outer electron further from nucleus, lost more easily
10	Why Group 7 gets less reactive down the group				Outer shell further from nucleus, electron gained less easily

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C1 Atomic Structure and the Periodic Table – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 6: Transition Metals Paper 1 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Position of transition metals				Central block, between Groups 2 and 3
2	Transition metals compared with Group 1				Harder, denser, higher melting point, less reactive
3	Examples of transition metals				Cr, Mn, Fe, Co, Ni, Cu
4	Example of ions with different charges				Fe^{2+} and Fe^{3+}
5	Appearance of transition metal compounds				Coloured
6	Common use of transition metals				Catalysts
7	Catalyst in the Haber process				Iron
8	Colour of copper(II) compounds				Blue
9	Colour of iron(II) compounds				Green
10	Colour of iron(III) compounds				Orange-brown

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.